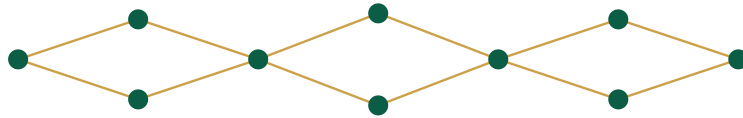


SAIF · KINGDOM OF SAUDI ARABIA

SANDOOQ

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Sovereign, end to end encrypted cloud storage, owned by the
people who use it.



1. The problem

I ran out of storage again.

There are only ever two options on the screen: delete photos I actually want, or pay for more space. I have chosen "pay" so many times that I stopped counting. And then one question started bothering me and would not go away. Why am I renting room on a computer I will never see, when I already own storage that sits there doing nothing?

That question became this project. Storage is the only utility that becomes more expensive the longer you depend on it. Every phone ships with exactly enough space to create demand, and the only remedy offered is a subscription that never ends. This is not a personal annoyance. It is a global structure that quietly moved the world's personal data onto someone else's balance sheet, and it did so while making renting feel normal.

1.1 Three stacked problems

1. **The capacity trap.** Your device fills up, and the paid upgrade is positioned as the only answer.
2. **The ownership gap.** You pay for the storage, but you never own it, and you can never switch it off.
3. **The sovereignty gap.** Your files are copied to data centres in America and other countries, under laws you never agreed to.

1.2 What actually happens to your files

Conventional cloud

- Files are copied to foreign data centres.
- The provider holds the encryption keys, so it can technically read everything.
- Privacy is a policy that can be rewritten overnight, not a fact you can test.
- You cannot audit who has access or where copies exist.
- Legal authority over your data sits in a foreign jurisdiction.
- Each month, the payment leaves the country permanently.

Sandooq

- Files are cryptographically sealed on the user's own device before they move.
- The operator never holds a key, so it cannot read anything, at any point.
- Privacy is mathematics, and mathematics does not change its mind.
- Anyone can verify it: the storage disk shows unreadable blocks only.
- Saudi data stays under Saudi rules, inside the Kingdom.
- Each month, the payment becomes income for a host inside the country.

1.3 The invisible half: wasted capacity

While millions of people pay for storage, millions more already own storage they never use. Old laptops, external drives and NAS boxes sit idle in homes, schools and small offices across the Kingdom, producing nothing, because there has never been a safe and simple way to share them or to earn from them. That unused capacity is the raw material of this project.

2. The solution

Sandooq is a self hosted, end to end encrypted cloud platform combined with a marketplace for storage. It does two things that no single mainstream product does at the same time:

- **It gives users sovereign storage.** Encrypted on the device, stored in the Kingdom, verifiable by the user, and reachable from any device.
- **It turns idle hardware into income.** Anyone with spare capacity can run a node and earn monthly revenue from the space they are not using.

The design principle: zero trust toward the operator. The system is built so that the company running it, me included, cannot read user files. Privacy is not a promise here, it is an architectural constraint.

3. The technology, in full detail

3.1 The applications used

Application	Role in the project
Termius (iPhone and Mac)	The secure terminal client. It connects to the server with an SSH key over the custom port 52156, so administration happens entirely over an encrypted channel.
macOS Terminal + ssh-keygen	Generates the Ed25519 key pair. The private key never leaves the device, and only the public key is shared with the server.
Apple Files app	The native file browser on iOS, used as the destination for the storage nodes through SMB and through file provider extensions.
Samba (SMB)	The file sharing protocol used for local network storage and for reaching shares from the iPhone.
Cryptomator (iPhone, macOS)	Creates the encrypted vault. Files are encrypted on the device before upload, so the server stores blocks it cannot read.
Nextcloud or Seafile	The cloud platform that becomes Sandoq: user accounts, quotas, sharing, mobile apps, a web interface and federation between nodes.
Tailscale	A private encrypted network that lets devices reach the node from anywhere without exposing ports. Tailscale Funnel can temporarily publish a service for a public demonstration.
Secure ShellFish or Owlfiles	File provider applications that bring SFTP storage into the native Files app, without giving up the SSH key model.
Let's Encrypt	Provides the free TLS certificate that gives the platform a trusted HTTPS connection.
Docker or snap	Packaging and deployment of the cloud platform on the server, in a repeatable, isolated way.

3.2 The protocols and ports

Protocol	Purpose	Port
SSH	Key based administration, and the tunnel that SFTP runs inside.	52156 (custom)
SFTP	Encrypted file transfer using the same keys.	inside SSH
HTTPS / TLS	Encrypts every web and app session and proves the identity of the platform.	443
SMB	File sharing for Mac, Windows and the iPhone Files app on a local network.	445
AFP	Apple's retired file protocol, deliberately not used.	deprecated
WebDAV	Web based file access for compatible applications.	80 / 443
DNS	Resolves the domain name to the node's public address.	53

3.3 The three layers of encryption

Layer	What it protects	How it is achieved
In transit	Data while it is moving	HTTPS and TLS for apps and web, SSH for administration and transfer
At rest	Data while it sits on a disk	Full disk encryption on the node, plus the encrypted vault itself
End to end	Data from everyone, including the operator	Cryptomator vaults, where the key is created on the user's device and never transmitted

4. The challenges I faced, and how I solved them

Challenge	How I solved it
Tooling friction. The older Termius build on the Mac had no visible key manager, and the server was not on the default SSH port, so standard tutorials did not apply.	Generated the Ed25519 key directly in the macOS Terminal with ssh-keygen , and used the custom port 52156 explicitly in every command. Both problems became deliberate security choices.
The iPhone Files app refused to connect to the server. I expected to type an address and browse files.	Discovered that the Files app only speaks SMB, not SSH or SFTP, and that AFP is retired. The fix was a design decision: give the Files app an SMB or file provider route, and keep SSH for the terminal.
A Mac share only worked on my own Wi Fi. Addresses like 192.168.1.11 are private and cannot be reached from outside.	Understood NAT and CGNAT, and moved the public role to the rented server, which has a real public address. The Mac became the rehearsal, the server became the product.
Cryptomator looked broken, twice. First I saw a photo on my laptop and assumed encryption had failed, then I discovered another photo had been saved next to the vault instead of inside it.	Learned that a vault protects data from the storage and from other people, not from the key holder, and that files are only protected when they are inside the vault. Verified it by opening the storage folder without unlocking and seeing scrambled names only.
The vault worked, but I had no way to prove privacy to anyone.	Built a repeatable demonstration: encrypt on the device, then show the storage folder and display only unreadable blocks. Proof beats a promise in front of judges.
Terminology was overwhelming. Encryption, TLS, SMB, SFTP, NAT, federation.	Wrote a bilingual glossary as I learned, so the concepts became usable rather than intimidating. It is now a companion document to this report.
The first network map looked wrong and put Riyadh at the centre. It undermined the idea of decentralisation.	Rebuilt the map from real province level geographic data, placed twelve equal peer nodes with no hub, and verified every node sits inside its own region.
Choosing an architecture. SMB, SFTP, a cloud platform, or all three.	Settled on a layered design: SMB and file providers for access, SSH for administration, and a self hosted cloud platform for accounts, quotas and federation.

5. Why Sandoog is better than iCloud

	iCloud	Sandoog
Who can read your files	Apple holds the keys. You trust a policy.	Only you. The key never leaves your device.
Where your files live	Data centres abroad.	Inside Saudi Arabia.
Can you verify privacy	No. You take their word.	Yes. The disk shows unreadable encrypted blocks.
What you pay	A subscription, forever.	A free tier, then plans priced below mainstream providers.
Who earns from your storage	A foreign corporation.	Saudi hosts, families and small businesses.
Economic effect	Money, data and skills leave the country.	Money, data and skills stay in the Kingdom.

In one line: iCloud rents you space on someone else's computer. Sandooq gives you encrypted space on infrastructure you and your neighbours own, and pays those neighbours for it.

6. The core idea: a marketplace, not just a server



1. **Install the host app** on a computer or NAS and connect it to Sandooq.
2. **Declare capacity and availability**, and pass a readiness check for uptime, encryption and speed.
3. **Go live.** The node appears in the app with its price, latency and trust signals.
4. **Serve users and get paid.** The host stores encrypted blocks it cannot read and receives a monthly payout for the space actually used.

7. A federated mesh across the Kingdom

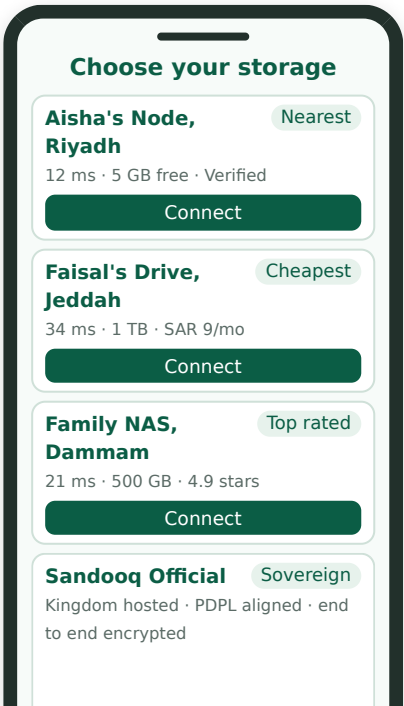
Sandoog is not one central server with smaller servers attached. It is a mesh of equal peer nodes, each independently operated, joined through federation, with no centre and no single owner. That is what makes it sovereign, resilient and fair.

A federated mesh. Equal peer nodes. No central server.

Anyone can run a node. Lines show federation, not hierarchy.

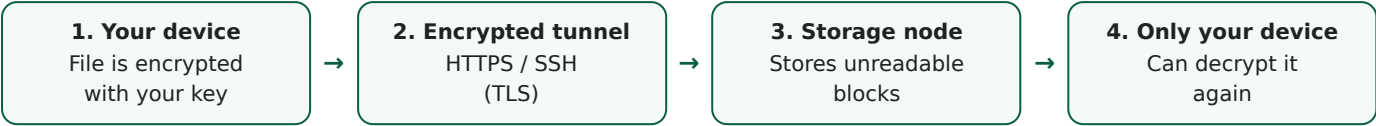


Every dot is a node an independent operator can run, from a student with one external drive to a small business with a rack. The lines are federation links, not a hierarchy. Users choose by price, latency and trust, and are never locked to one operator.

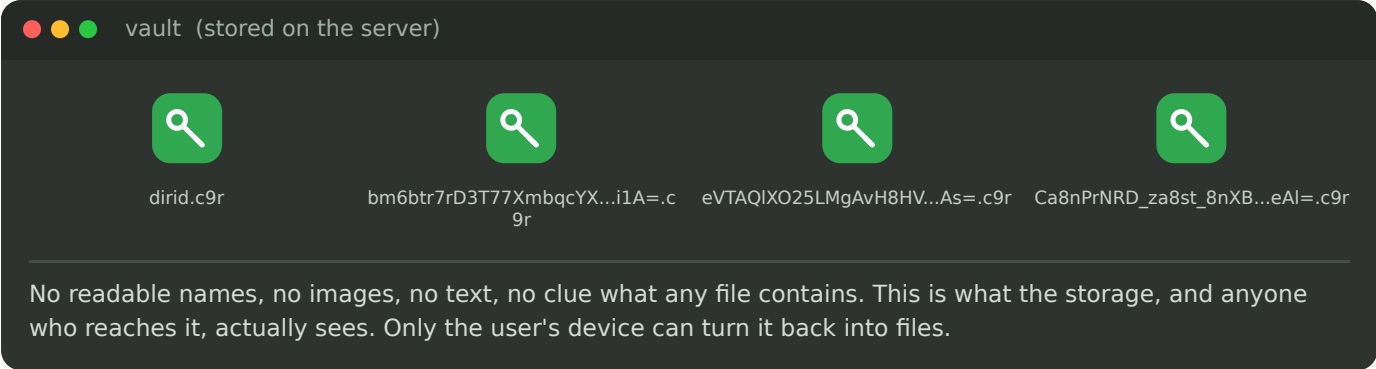




8. Privacy, cryptographically



This is the actual content of the vault on the storage:



Who	Can read the files?	What they can see
The user	Yes, they hold the key	Everything they own
Node operator or host	No	Encrypted blocks, file counts, rough sizes, activity
Other users	No	Only what is explicitly shared
Network eavesdropper	No	Scrambled traffic
Thief with the disk	No	Unreadable storage

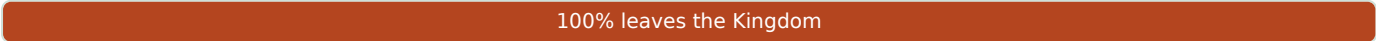
Stated plainly: a host can still observe metadata, and anyone controlling a disk can always delete data. What no one can do is read it. That claim is testable, and users are invited to test it.

9. The economics

Item	Example
Host offers	500 GB on an idle external drive
Used in month one	40% = 200 GB rented
Market price	SAR 10 per 100 GB per month
Gross revenue	SAR 20 per month
Platform commission (20%)	SAR 4
Host payout	SAR 16 per month , from hardware that was earning nothing

Illustrative figures. They show the shape of the model, not a promise of income.

Today: a subscription to a foreign provider



Sandoog: the same payment, redistributed locally



● stays with Saudi hosts ● platform commission ● leaves the country

Potential business model

Stream	How it works
Subscriptions	A free starter tier, then paid plans in SAR priced below mainstream providers.
Marketplace commission	A percentage of every rental between a user and a host, growing automatically with the network.
Family and group plans	A shared pool for families and friend groups: one payer, many users.
Schools and small businesses	Private, PDPL aligned storage with data residency in the Kingdom.
Managed hosting and white label	Setup and maintenance of nodes for organisations, and licensing to local brands and ISPs.

10. Market, rollout and risks

First users: students with full phones and no budget, families with scattered photos, privacy focused users, small businesses that need local storage, and hosts who want their idle hardware to pay for itself.

Channels: short form video that shows a photo turning into unreadable data; campus ambassadors; gaming and creator communities; national technology events such as LEAP and SDAIA initiatives; device retailers and ISPs as host onboarding points; a referral programme; and plain Arabic and English education content.

Phase	Goal
0. Proof of concept (now)	One working node, HTTPS, accounts, quotas, an encrypted vault, and a live host rental demo.
1. Private pilot (1 to 3 months)	10 to 30 testers, onboarding, backups, monitoring.
2. Multi user product (3 to 6 months)	Self service signup, two factor, mobile apps, billing, support.
3. Federation (6 to 12 months)	Multiple nodes, a node directory, and the in app node picker.
4. Market launch (12 to 18 months)	Host app, reputation, payouts, commission.
5. Scale (18 months and beyond)	A national network, B2B and white label, partnerships, compliance programme.

Risk	Response
Security attacks	HTTPS everywhere, key only SSH, updates, firewall, two factor, least privilege.
Abuse and data loss	Clear terms, account limits, a take down process, and backups in a second location with tested restores.
Untrustworthy hosts	End to end encryption removes the ability to snoop; verification tiers and public reputation manage reliability.
Regulation	In Kingdom hosting, documented policy, minimal metadata, and clear consent under PDPL.

Sandooq. Storage you own. Privacy you can verify. Income that stays in the Kingdom.

Saudi based, self hosted and encrypted end to end, and built so that the people who provide the storage are the people who earn from it.

